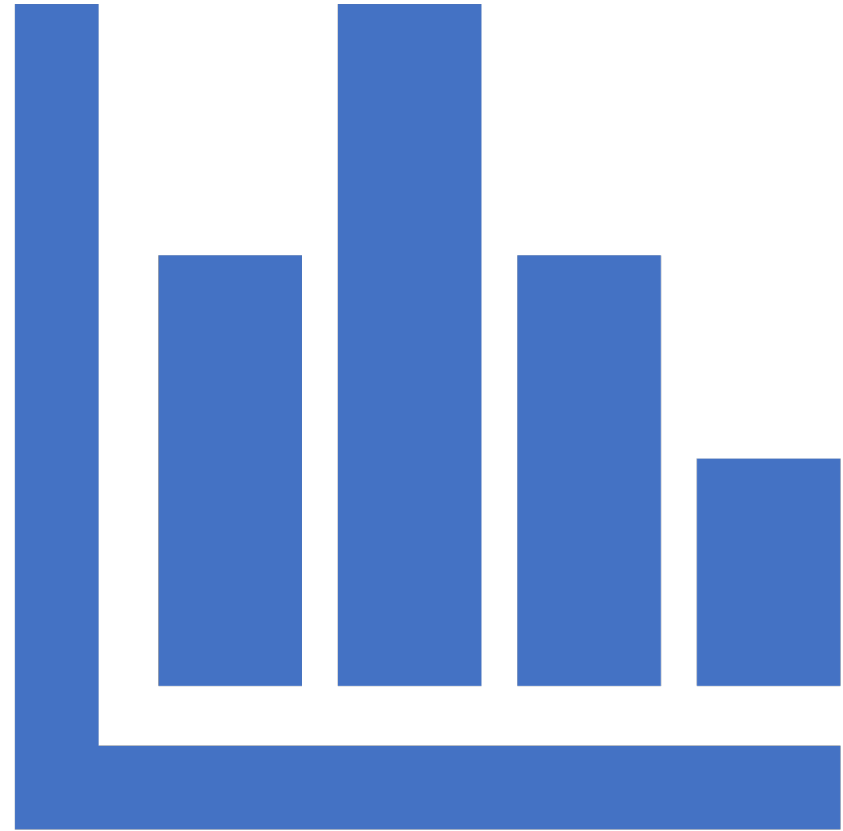


Examples: Distributions of
Quantitative Variables

Stat 3113: Lesson 5

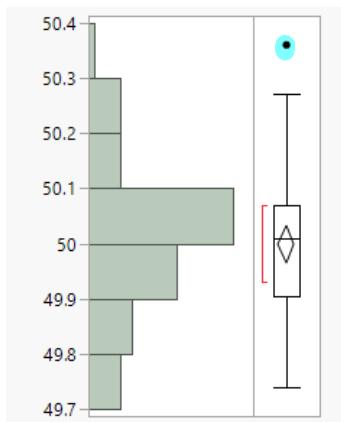


Example 1:

- In order to investigate the actual probability distribution of the diameters of the cylinders that a company produces, the company selects a random sample of 60 cylinders and measures their diameters.
- The population may be the set of all cylinders produced by the company (within a certain time period) or, if attention is directed at only one production line, say, the population may be all cylinders produced from that production line. The random selection of the 60 cylinders that constitute the sample should ensure that it is a representative sample.
- Excel data file called **diameters** is on Canvas.

50.08	50.05	49.81	50.17	50.00	49.95
49.78	49.94	50.02	50.12	50.03	50.20
50.02	50.19	50.26	50.00	49.92	50.03
50.02	49.86	49.90	50.01	50.07	49.92
50.13	50.03	50.01	49.85	49.89	50.02
49.74	50.04	50.04	49.93	49.99	49.97
49.84	49.96	50.01	49.84	50.01	50.27
49.97	49.90	49.79	50.20	50.09	49.77
49.93	49.87	50.36	49.94	49.90	50.07
50.02	50.13	50.21	49.74	50.05	50.07

Using JMP



Quantiles		
100.0%	maximum	50.360001
99.5%		50.360001
97.5%		50.312751
90.0%		50.199001
75.0%	quartile	50.07
50.0%	median	50.009998
25.0%	quartile	49.905001
10.0%		49.813001
2.5%		49.740002
0.5%		49.740002
0.0%	minimum	49.740002

Summary Statistics

Mean	49.998833
Std Dev	0.1336968
N	60
Minimum	49.740002
Maximum	50.360001
Median	50.009998
Interquartile Range	0.164999

Example 1: Let's do the five aspects of the distribution and discuss.



Outliers: $\gamma_5 \rightarrow x = 50.36001$



Center:
 • Mean ($\bar{x}$) $\rightarrow 49.998833$
 • Median ($\tilde{x}$) $\rightarrow 50.009998$



Spread:
 • $s \rightarrow 0.1336968$
 • IQR $\rightarrow 0.164999$



#groups: 1 group $\rightarrow$ Unimodal

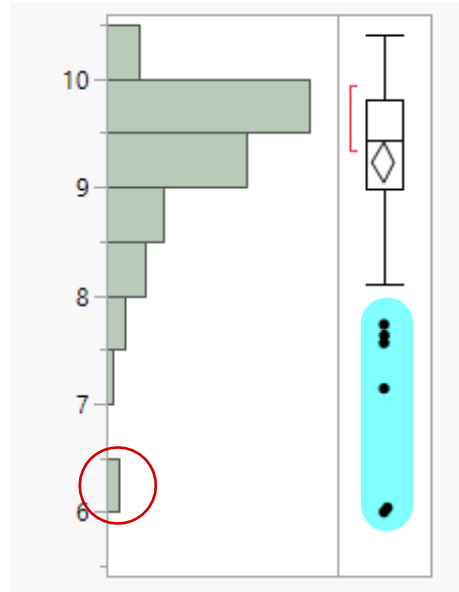


Symmetry: Approximately symmetric



Example 2:

- The bending capabilities of plastic panels are investigated by measuring the angle of bend at which a deformity first appears in the panel.
- A researcher collects 80 plastic panels made by a machine and measures their deformity angles.
- Excel data file called ***bending*** is on Canvas.



Quantiles		
100.0%	maximum	10.399999619
99.5%		10.399999619
97.5%		10.294250178
90.0%		9.9300003052
75.0%	quartile	9.8100004196
50.0%	median	9.434999428
25.0%	quartile	8.9824995995
10.0%		8.1219999313
2.5%		6.0674999595
0.5%		6
0.0%	minimum	6

Summary Statistics	
Mean	9.229375
Std Dev	0.8422789
N	80
Minimum	6
Maximum	10.4
Median	9.4349999
Interquartile Range	0.8275008

Example 2: Let's do the five aspects of the distribution and discuss.

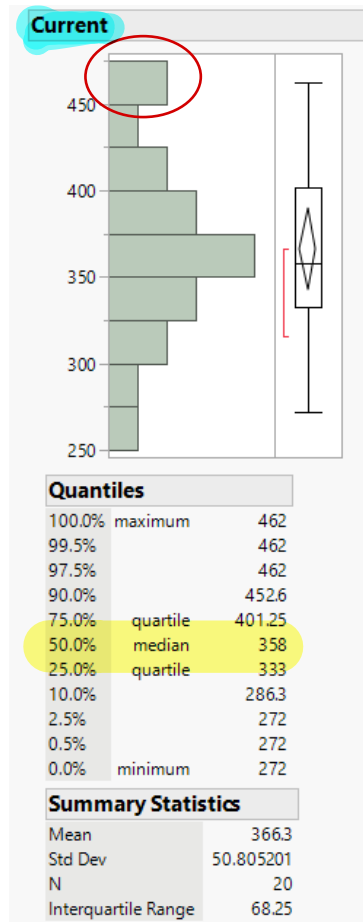
- **Outliers:** *yes → About 6 outliers in the lower tail*
- **Center:**
 - Mean ($\bar{x}$) → *9.229375*
 - Median ($\tilde{x}$) → *9.4349999*
- **Spread:**
 - *s* → *0.8422789*
 - IQR → *0.8275008*
- **#groups:** *1 group → Unimodal*
- **Symmetry:** *left-skewed*



Example 3:

- Yummy Chickens company is trying to find a way to fatten their baby chicks up faster so they can get more money for them at market. They believe a new type of feed might accomplish this. The cost of the new feed is the same as the current feed they are using.
- A sample of 40 baby chicks of the same age are selected. The 40 chicks are randomly separated into two separate groups. Group 1 will be fed the current corn and group 2 will be fed the new corn for a two-month period. The weight gain in ounces is measured to be analyzed. The Excel data is on Canvas entitled **corn**.
- Let's go through the five aspects for both samples and answer the question, does it appear the new corn is resulting in significant weight gain?

Example 3:



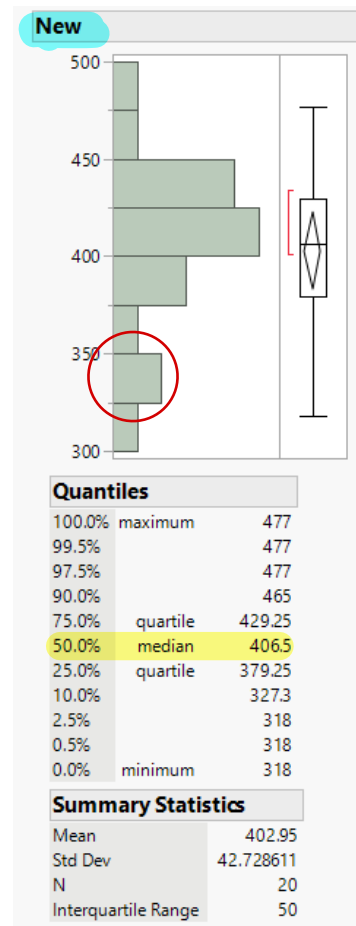
Outliers: *No Outlier*

Center: • Mean ($\bar{x}$) → 366.3
• Median ($\tilde{x}$) → 358

Spread: • s → 50.805202
• IQR → 68.25

#groups: 1 group → Unimodal

Symmetry: *Approximately Symmetric*



Outliers: *No Outlier*

Center: • Mean ($\bar{x}$) → 402.95
• Median ($\tilde{x}$) → 406.5

Spread: • s → 42.728611
• IQR → 50

#groups: 1 group → Unimodal

Symmetry: *Approximately symmetric*

Example 3: Final answer with justification.

- It appears that the new corn is resulting in significant weight gain.
- Justification
 - Compare the means first.
 - Then the medians to cast away any doubt.

Reading

Chapter 1 – Section 1.2, 1.3

Next Topic



Introduction to Probability Part 1



Chapter 3 – Section 3.1